Effects of Combined Oral Contraceptive Ethinylestradiol (30 µg) and Dienogest (2 mg) on Carbohydrate Metabolism During 1 Year of Conventional or Extended-Cycle Use

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Key words
- combined oral contraceptives
- extended-cycle regimen
- glucose metabolism
- insulin
- C-peptide
- glucose tolerance

Abstract

The effects of extended regimens of combined oral contraceptives (COCs) on carbohydrate metabolism are largely unknown. The present study compared the effects of a COC containing 30 µg ethinylestradiol and 2 mg dienogest (EE/DNG) in conventional and extended-cycle regimen over 1 year. Parameters of carbohydrate metabolism were measured in 59 women treated with EE/DNG either conventionally (13 cycles of 21 + 7 days) or in extended-cycle regimen (4 cycles of 84 + 7 days). Blood samples were taken in a control cycle, and at 3 and 12 months of treatment. The mean levels of HbA1c and fasting glucose levels remained stable in both conventional and extended-regimen of EE/DNG. The mean levels of fasting insulin and C-peptide underwent comparable increases in both regimens, suggesting a similar readjustment of glucose metabolism via slightly increased insulin secretion. For both regimens, the response to the oral glucose tolerance test (OGTT) showed a slightly impaired glucose tolerance and insulin resistance at 3 months. These changes improved or returned to baseline at 12 months. Accordingly, the mean index for insulin resistance (homeostasis model assessment of insulin resistance, HOMA-IR) increased and the mean insulin sensitivity index [ISI (composite)] decreased modestly in both groups. The present study demonstrates that there are no statistically significant differences between the effects of conventional and extended-cycle treatment on carbohydrate metabolism over 1 year of treatment. In general, the effects of both regimens were moderate and mostly transient.

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Introduction

Modern combined oral contraceptives (COCs) are known to increase plasma glucose response by 10–50% during oral glucose tolerance tests (OGTT) [1–4]. Also, extreme OGTT responses are twice as frequent in COC users compared to non-users, which may be a health concern [5]. This effect is attributed to the estrogen component, since all low-dose COCs cause a similar degree of insulin resistance. In addition to the estrogen effects, the type and dose of the progestogen component may modulate the elimination of insulin from plasma [4].

A number of women experience various somatic and psychic complaints during the cyclic hormonal fluctuations [6, 7]. During pill intake the concentrations of estrogen and progestogen rise to a steady state and rapidly fall to baseline levels in the hormone-free week [8]. Accordingly, the serum levels of many hormone-sensitive proteins and metabolic factors undergo cyclic fluctuations. To avoid cycle-dependent symptoms or menses-related disorders, many affected women omit the hormone-free interval and, indeed, international surveys indicate that a majority of women would prefer extended regimens with few bleeding episodes instead of conventional bleeding patterns every 4 weeks [6, 9].

Although many advantages of the extended-cycle regimens are obvious or probable, data concerning their risks are still limited. Since the safety of new COC formulations can be assessed only after several years of use in a sufficiently high number of women, a preliminary estimation of health risks requires the investigation of surrogate safety parameters.

The present study focused on the investigation of parameters of carbohydrate metabolism. These were measured in a subgroup of 59 women at the Center of Obstetrics and Gynecology, University Hospital of Frankfurt, within the frame of a large multi-center, comparative, prospective, randomized, 12-month study on efficacy and safety.

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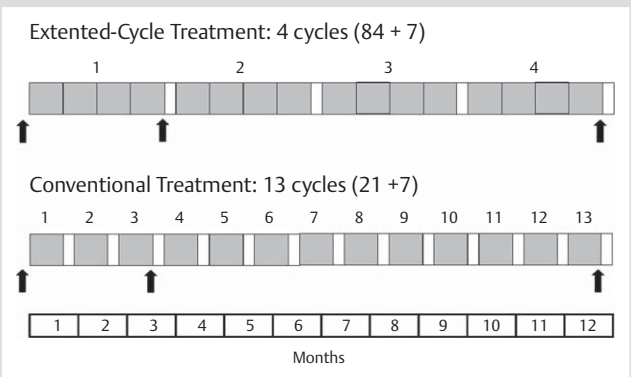

Fig. 1 Flow chart. Arrows indicate blood sampling schedule during the control cycle and at 3 and 12 months of treatment. White bars indicate hormone-free interval of 7 days.

The effects of conventional treatment (21+7 days) with a combination of 30 µg EE and 2 mg DNG (EE/DNG) on the serum concentration of HbA1c, glucose, insulin, and C-peptide after overnight fasting and after glucose load were compared with those of an extended-cycle regimen (84+7 days) with EE/DNG. The mean indices for insulin resistance (homeostasis model assessment of insulin resistance, HOMA-IR) and insulin sensitivity [ISI (composite)] were also calculated. Measurements were carried out at baseline and at 3 and 12 months of treatment.

Materials and Methods

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Design of the study

Screening of 66 healthy women between 18 and 40 years of age seeking contraception resulted in inclusion of 60 volunteers with regular menstrual cycles and without contraindications for the use of COCs. The reasons for the 6 screening failures were: obesity (body mass index >30 kg/m^2), thalassemia minor, post-traumatic syndrome, Hashimoto's thyroiditis, laboratory values outside of inclusion range, and withdrawal of consent. In addition, one volunteer of the conventional-regimen group became pregnant in the control cycle and did not receive any study medication, leading to a group size of n=29.

The participants had not used any hormonal medication for at least 4 weeks prior to the study and did not use drugs known to influence the effects of OCs. The volunteers did not use any drugs known to influence the action of insulin for at least 4 weeks prior to blood sampling, except one volunteer of the conventional group who took once 40 mg methylprednisolone one day before the OGTT in week 51, and another volunteer of the conventional group who was treated for hypothyroidism with 50 µg levothyroxine per day starting from week 25.

After the control cycle, the volunteers were randomly assigned to take a monophasic COC containing 30 µg EE and 2 mg DNG (EE/DNG) either conventionally (13 cycles with 21 days of treatment+7 days without hormones) (n=29) or according to an extended-cycle regimen (4 extended cycles with 84 days of continuous administration + 7 days without hormones) (n=30) (**Fig. 1**). The randomization was performed using randomization blocks by means of the software package SAS. The random numbers were equally distributed to the two treatment groups. The allocation of a randomization number was chronologically linked to the sequence of arrival of the volunteers on Visit 2. Neither the volunteer nor the investigator knew the identity of the study regimen before assignment of the random numbers.

Blood was taken on Days 21–26 of the control cycle and on Days 82–84 of the 1st and 4th extended cycle (3 and 12 months of treatment, respectively); or on Days 19–21 of the 3rd and 13th conventional cycle (3 and 12 months of treatment, respectively) (**Fig. 1**). Blood sampling was carried out in the morning in a lying position after overnight fasting for at least 12 h.

In addition to the central approval of the Ethics Committee for the whole study (Medical Council of the German Federal State of Thuringia), this part of the study was approved by the Ethics Committee of the University Hospital of Frankfurt and was conducted in compliance with Good Clinical Practice (GCP) and the Declaration of Helsinki. Each volunteer signed a written informed consent before enrollment in the study. The study was conducted between June 2003 and February 2005.

Laboratory methods

The carbohydrate metabolism was evaluated by determination of HbA1c and by a 75 g standardized oral glucose tolerance test (OGTT) in a control cycle and at 3 and 12 months of treatment. The measurements were performed centrally by the laboratory LKF GmbH, Kiel, Germany.

HbA1c was measured in EDTA-blood stored at room temperature until analysis, performed by high performance liquid chromatography (BioRad Variant™ Hemoglobin A1c, Bio-Rad Laboratories, Munich, Germany) (intra-assay CV 0.7–1.0%; interassay CV 3.05–3.14%; sensitivity 2.1%).

The OGTT consisted of 5 blood samples obtained within 30 min before glucose ingestion, and 30, 60, 90, and 120 min thereafter for the determination of glucose, insulin, and C-peptide. Blood samples were taken in heparin monovettes, cooled (+4 °C) until centrifugation, and centrifuged within 30 min at +4 °C for 15 min (2 500 revolutions per minute). Plasma (1 ml) was transferred into a plasma tube and sent to the laboratory for glucose determination. For the determination of insulin and C-peptide, 1 ml of plasma was transferred into a plasma tube and stored at –70 °C until dry ice transport to the laboratory.

The plasma glucose concentrations were determined using the hexokinase method (Gluco-quant® Glucose HK, Roche Diagnostics, Mannheim, Germany) [intra-assay coefficient of variability (CV) 1.0%; interassay CV 1.7%; sensitivity 0.11 mmol/l]; of insulin by a direct solid-phase chemiluminescence immunoassay (Elecsys® Insulin Immunoassay, Roche Diagnostics) (intra-assay CV 1.9%; interassay CV 2.6%; sensitivity 0.2 mU/l); and of C-peptide by a double-antibody radioimmunoassay (EURO/DPC's Double Antibody C-Peptide, DPC Biermann, Bad Nauheim, Germany) (intra-assay CV 3.4%; interassay CV 10.0%; sensitivity 0.22 ng/ml).

The homeostasis model assessment of insulin resistance (HOMA-IR) was calculated using the formula: fasting insulin (mU/l) × fasting glucose (mmol/l)/22.5 [10].

The composite whole body insulin sensitivity index during the OGTT [ISI (composite)] was calculated using the formula: 10 000/square root of [fasting glucose (mg/dl) × fasting insulin (mU/l)] × [mean OGTT glucose concentration (mg/dl) × mean OGTT insulin concentration (mU/l)] [11].

Statistical analysis

The sample size was established as 60 subjects (30 in each study arm). This sample size was chosen without statistical consideration, because it is commonly accepted by regulatory authorities

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for this type of study. Analyses were based on the full analysis set (FAS). The FAS was defined according to the intention-to-treat principle, restricted to the volunteers who had taken at least one tablet of the study medication and had at least one observation. Missing data were not replaced.

For HbA1c, fasting glucose, fasting insulin, fasting C-peptide, and the AUC results, the group differences were evaluated by means of analysis of variance (ANOVA) for repeated measures, with a significance level set at $p \leq 0.05$. In the case of the AUC results, analyses were based on logarithm-transformed values.

For HOMA-IR and ISI (composite), group comparisons were carried out by Students's *t*-test, with a significance level set at $p \leq 0.05$.

Results

Disposition of volunteers

Sixty volunteers were randomized to be treated with EE/DNG either for 13 conventional cycles (21+7 days) or 4 extended cycles (84+7 days). One volunteer became pregnant in the control cycle and dropped out before the first medication, reducing the conventional group to n=29. Two women prematurely discontinued the study during the first extended cycle because of irregular bleeding and tinnitus, respectively. Consequently, 57 subjects completed this part of the study (conventional n=29, extended cycle n=28). In one volunteer of the extended-cycle group, a gastrointestinal infection precluded the performance of the OGTT and Hb1Ac at 12 months, resulting in a group size of n=27 at this time point.

The conventional and extended-cycle groups were comparable in terms of baseline data for age and body mass index (**Table 1**). The latter did not substantially change during the study, as observed also for blood pressure and pulse rate.

Fasting levels of HbA1c, glucose, insulin, and C-peptide

The mean levels of HbA1c were only minimally elevated in both groups and at both time points (3 and 12 months) compared to baseline (**Table 2**). Fasting glucose levels remained stable over time in both treatment groups (**Table 2**). Fasting insulin underwent a modest increase at 3 and 12 months compared to baseline, however, these changes were fully comparable in the two treatment groups.

For C-peptide, both conventional and extended cycles were accompanied by a modest and comparable increase after 3 and 12 months (**Table 2**). Thus, under fasting conditions the parameters of glucose metabolism did not show statistically significant differences between the conventional and the extended-cycle regimen at any time point.

Table 1 Characteristics of the study groups at screening

Characteristics	Conventional regimen (FAS; n=29)	Extended-cycle regimen (FAS; n=30)
Age (years)	24.6±3.2	25.1±2.6
Weight (kg)	63.7±8.0	62.0±8.1
Body mass index (kg/m^2)	22.3±2.8	21.8±2.6
Systolic blood pressure (mmHg)	109.1±10.9	112.2±10.6
Diastolic blood pressure (mmHg)	71.0±7.2	69.8±9.1
Heart rate (beats/min)	71.9±7.6	70.9±7.9

Values are mean±SD. FAS: Full analysis set

Table 2 Effects of conventional or extended-cycle use of a monophasic combination of 30 µg ethinylestradiol and 2 mg dienogest (EE/DNG) on parameters of glucose metabolism after 3 and 12 months of treatment, compared to the control cycle

Parameter	Conventional regimen Absolute values			Changes over time		Extended-cycle regimen Absolute values			Changes over time	
	Baseline (n=29)	3 months (n=29)	12 months (n=29)	3 months (n=29)	12 months (n=29)	Baseline (n=30)	3 months (n=28)	12 months (n=27)	3 months (n=28)	12 months (n=27)
HbA1c (%)	4.8±0.4	5.0±0.3	5.0±0.3	0.1±0.2	0.2±0.2	4.8±0.4	4.9±0.4	4.9±0.3	0.1±0.3	0.1±0.3
Glucose (mmol/l)	4.9±0.3	4.9±0.4	4.8±0.4	0.01±0.4	−0.16±0.4	5.1±0.4	4.9±0.4	5.0±0.4	−0.1±0.5	−0.07±0.4
Insulin (mU/l)	5.9±2.4	8.0±3.8	7.6±4.2	2.1±3.7	1.8±3.6	6.7±3.2	8.2±3.7	7.5±2.9	1.6±3.2	1.0±3.4
C-Peptide (ng/ml)	1.6±0.5	1.8±0.4	1.8±0.5	0.2±0.5	0.3±0.5	1.7±0.5	1.9±0.6	1.9±0.6	0.2±0.4	0.2±0.7
$AUC_{glucose}$ (mmol/l*min)	686±136	736±155	668±111	50±153	−18±109	685±150	747±139	743±142	69±144	64±157
$AUC_{insulin}$ (mU/l*min)	5084±2306	6337±3216	5061±2615	1253±2697	−23±1970	5865±3250	6541±4680	5941±3398	819±2264	287±2086
$AUC_{C\ peptide}$ (ng/ml*min)	660±184	751±275	654±215	91±225	−6±155	710±240	735±219	705±252	26±165	4±191

Values are mean±SD. The upper part of the table presents the fasted values of HbA1c, glucose, insulin, and C-peptide. The lower part presents the mentioned parameters from OGTT over 120 min

AUC: Area under the (concentration) curve. Due to drop-outs the changes over time in the extended-cycle regimen group refer to the remaining 28 and 27 volunteers, respectively

None of the differences between the conventional and extended-cycle regimen were statistically significant (ANOVA for repeated measures)

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Table 3 Effects of conventional or extended-cycle use of a monophasic combination of 30 µg ethinylestradiol and 2 mg dienogest (EE/DNG) on parameters of glucose metabolism during OGTT at baseline and after 3 and 12 months of treatment, compared to the control cycle

Parameter	Conventional regimen					Extended-cycle regimen				
	−30 min	+30 min	+60 min	+90 min	+120 min	−30 min	+30 min	+60 min	+90 min	+120 min
Glucose (mmol/l)										
Baseline	4.94±0.31	6.92±1.43	5.94±1.76	5.14±1.37	4.82±1.13	5.08±0.42	7.03±1.37	5.74±2.02	5.09±1.40	4.90±1.06
3 months	4.95±0.41	7.28±1.36	6.43±1.99	5.73±1.59	5.25±1.43	4.94±0.40	7.76±1.44	6.36 ±1.68	5.71±1.51	5.18±1.18
12 months	4.78±0.42	6.81±1.19	5.72±1.42	4.97±1.00	4.76±0.88	4.98±0.40	7.34±1.09	6.49±1.81	5.81±1.78	5.27±1.42
Insulin (mU/l)										
Baseline	5.87±2.43	52.58±22.73	52.57±27.69	44.70±29.00	33.36±15.17	6.72±3.21	68.37±51.67	56.75±29.28	48.05±33.47	37.94±21.32
3 months	7.97±3.82	62.36±28.59	66.99±37.45	55.75±34.73	44.27±31.20	8.15±3.71	72.92±59.29	65.80±57.89	55.09±39.64	40.31±21.44
12 months	7.62±4.23	54.10±26.72	51.73±31.89	42.84±30.90	32.45±17.10	7.54±2.86	61.26±41.61	57.87±33.45	56.01±43.95	38.23±22.46
C-peptide (ng/ml)										
Baseline	1.58±0.49	5.36±1.29	6.63±1.98	6.51±2.79	5.44±1.51	1.70±0.47	6.28±2.93	6.90±2.36	6.63±2.73	5.44±1.51
3 months	1.80±0.42	5.90±1.91	7.62±3.09	7.32±3.41	6.57±3.12	1.91±0.57	6.25±2.27	7.15±2.21	7.02±2.30	6.24±2.17
12 months	1.84±0.55	5.28±1.81	6.66±2.52	6.18±2.47	5.52±1.96	1.91±0.64	5.83±2.90	6.87±2.48	7.00±3.13	5.67±1.96

Values are mean ± SD
OGTT: oral glucose tolerance test

Oral glucose tolerance test (OGTT)

At 3 months, the mean glucose response was slightly increased in both groups (**Table 2, 3**). At 12 months, the mean values returned to baseline in the conventional regimen group, but remained slightly elevated in the extended-cycle group (**Table 2**), although this group difference was not statistically significant.

The mean insulin response was modestly increased after 3 months of conventional and extended use, returning approximately to baseline levels at 12 months in both treatment groups (**Table 2, 3**). Similarly, the response of C-peptide was only minimally increased in both groups at 3 months, returning to baseline levels at 12 months in both cases (**Table 2, 3**).

Thus, under glucose loading conditions the parameters of glucose metabolism did not show statistically significant differences between the conventional and the extended-cycle regimen at any time point.

Surrogate indices for insulin resistance and insulin sensitivity

After 3 months of treatment, the mean index for insulin resistance (HOMA-IR) was increased by approximately 40% in the conventional group and by 20% in the extended-cycle group. At 12 months the HOMA-IR remained modestly increased in both groups. In terms of mean insulin sensitivity index [ISI (composite)], at 3 months this was reduced by approximately 20% in both groups. At 12 months, this reduction was still present in the extended-cycle group, whereas it was reversed in the conventional group (**Table 4**). At any rate, there were no significant group differences, whether at baseline, 3, or at 12 months of treatment.

Discussion

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This randomized study directly compared the effects of conventional and extended use of EE/DNG on carbohydrate metabolism over 1 year. Notably, at 3 and 12 months of treatment there were no statistically significant differences between the two regimens for any of the parameters, whether under fasting conditions or during a standard challenge with oral glucose.

Both groups exhibited stable levels of HbA1c and fasting glucose compared to baseline. As glycosylated hemoglobin represents a retrospective indicator of the mean glucose concentration in blood over a prolonged period of time, the results suggest that glucose levels remain stable with both the conventional and the extended-cycle of EE/DNG.

The present results are comparable with those of previous studies showing no or modest effects on fasting glucose levels during treatment with 20–35 µg EE in combination with different types and doses of progestogens [12–20]. A modest but significant increase in fasting glucose (approximately 10%), in contrast, has been reported with COCs containing EE and desogestrel (DG) or gestodene (GSD) [21, 22], while a decrease was observed after 1 year of treatment with low-dose EE/levonorgestrel (LNG) [23].

To our knowledge, to date there is only one study available on the effects of an extended COC regimen on carbohydrate metabolism (continuous use of 30 µg EE and 75 µg GSD for 168 days in 45 women). This noncomparative study showed no change in fasting glucose, but a significant rise by 23% in fasting insulin [24].

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Table 4 Effects of conventional or extended-cycle use of a monophasic combination of 30 µg ethinylestradiol and 2 mg dienogest (EE/DNG) on HOMA-IR and ISI (composite) after 3 and 12 months of treatment, compared to the control cycle

Parameter	Conventional regimen			Extended-cycle regimen		
	Baseline (n = 29)	3 months (n = 29)	12 months (n = 29)	Baseline (n = 30)	3 months (n = 28)	12 months (n = 27)
HOMA-IR	1.29 ± 0.53	1.79 ± 0.98	1.62 ± 0.87	1.52 ± 0.74	1.81 ± 0.88	1.68 ± 0.72
ISI (composite)	8.65 ± 4.60	6.70 ± 3.69	8.09 ± 3.76	7.92 ± 4.46	6.59 ± 3.14	6.61 ± 2.46

Values are mean ± SD

HOMA-IR: homeostasis model of insulin resistance; ISI (composite): insulin sensitivity index

None of the differences between the conventional and extended-cycle regimen were statistically significant (Student's *t*-test)

In the present study, both regimens of EE/DNG caused a modest increase in fasting insulin at 3 and 12 months. Enhancement of insulin secretion is known to occur as physiological compensation for raised glucose levels [19], as also reflected by the slightly elevated C-peptide levels. Most studies with low-dose COCs report a significant increase in fasting C-peptide and no significant change in fasting insulin [12, 14–16, 18, 19, 23] or no significant effects on either parameter [17, 21].

At 3 months of conventional use of EE/DNG, the OGTT disclosed a slightly impaired glucose tolerance which, however, was reversed at 12 months (**Table 2**). The transiently elevated glucose response was accompanied by a transient, compensatory rise in insulin and C-peptide. In the extended-cycle group a slight deterioration of glucose tolerance was observed during the whole treatment period, although at 12 months neither the insulin nor the C-peptide changes were altered compared to baseline or were significantly different from those of the conventional regimen. Usually, only elevations of glucose levels combined with elevated insulin response are believed to indicate insulin resistance [19]. Before treatment, the mean initial insulin response (within 30 min) was 25 % higher in the extended-cycle group than in the conventional group. The difference, however, was largely due to 2 volunteers with high insulin levels 30 min after glucose load (260 mU/l and 186 mU/l) in the extended-cycle group. As the time-course of C-peptide suggested no difference in insulin secretion, the effect might be caused by a reduced hepatic insulin clearance in these individual volunteers.

For both EE/DNG treatment regimens, importantly, neither the mean fasting levels of glucose nor the mean glucose response to OGTT reached the limits of impaired glucose homeostasis set by the Expert Committee of the American Diabetes Association, that is, fasting plasma glucose of > 6.1 mmol/l (corresponding to > 110 mg/dl) and < 7.0 mmol/l (corresponding to < 126 mg/dl) and glucose levels between > 7.8 and < 11.1 mmol/l (corresponding to > 140 and < 200 mg/dl) 2 h after glucose load [25]. Notably, these thresholds were not exceeded in any of the individual volunteers.

In general, the estimation of HOMA-IR and ISI [composite] is useful for assessing insulin resistance and insulin sensitivity in interventional trials [26]. The present results of these indices support that the effect of the extended-cycle regimen on carbohydrate metabolism does not significantly differ from that of conventional treatment. In spite of the practical feasibility of these assessments in clinical trials, it must be kept in mind that clamp studies remain the gold standard methods for the quantification of insulin resistance [26].

Altogether, the slight and transient reduction in insulin sensitivity and glucose tolerance induced by both regimens of EE/DNG is similar to the changes observed during the luteal phase of spontaneous ovulatory cycles [27], and can therefore be considered clinically irrelevant.

The marginal influence on carbohydrate metabolism of conventional and extended treatment with EE/DNG may result from the lack of androgenic activity of DNG, a progestin with pronounced antiandrogenic properties. Similar effects on the response to OGTT have been also observed under a combination of 35 µg EE with cyproterone acetate, another potent antiandrogenic progestin [28].

It has been suggested that the estrogen component of COCs causes insulin resistance, while the progestins may induce a change in the pancreatic secretion and hepatic clearance of insulin [4, 29]. Previous studies have shown that impaired glucose tolerance occurs primarily in formulations containing progestins with androgenic activity [29]. The performance of OGTT in women treated with various low-dose COCs containing LNG, norethisterone (NET), or DG for at least 3 months was accompanied by a rise in the glucose AUC by 40–60 %, whereas a significant increase in the insulin response occurred only with combinations of 30 µg EE with 150 µg or 250 µg LNG [12, 19]. In contrast, no changes in the response to glucose load were observed using triphasic formulations with EE and LNG or DG and in a study with 30 µg EE and 150 µg DG [16, 22].

The present study clearly demonstrates that there are no statistically significant or clinically relevant differences between the effects of conventional and extended-cycle treatment on carbohydrate metabolism over 1 year of treatment. In general, the effects of both regimens were moderate and mostly transient.

Conclusion

Conventional or extended-cycle treatment with a combined oral contraceptive containing 30 µg ethinylestradiol and 2 mg dienogest caused only moderate changes in carbohydrate metabolism. There were no significant differences between the effects of conventional and extended regimens.

Conflict of Interest

The authors Inka Wiegratz and Herbert Kuhl received honoraria for lectures and expert reports from Jenapharm GmbH & Co. KG. The authors Katrin Mittmann, Ernesta Palombo-Kinne, Uwe Mellinger, and Evelyn Lange are employees of Jenapharm GmbH & Co. KG.

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